Classification Model Evaluation Report

Breast Cancer Diagnosis Prediction

Maincrafts Internship Task 4

1 Objective

This task shifts from regression to binary classification. A Logistic Regression model was trained on the Breast Cancer dataset to classify tumors as malignant or benign, evaluated using metrics beyond accuracy, and compared against a Decision Tree classifier. Class imbalance handling was also applied and assessed.

2 Methodology

- **Dataset**: scikit-learn's built-in Breast Cancer dataset (569 samples, 30 features). Target: malignant (0) vs. benign (1).
- **Split**: 80/20 train-test, `random_state=42`, **stratified** to preserve class distribution.
- **Preprocessing**: Features scaled with `StandardScaler`.
- **Baseline model**: Logistic Regression (`max_iter=1000`).
- **Imbalance handling**: Logistic Regression retrained with `class_weight="balanced"`.
- **Comparison model**: Decision Tree Classifier.

3 Metric Selection

Accuracy alone is misleading on imbalanced datasets, since a model that always predicts the majority class can still score high. Instead this report relies on:

- **Precision**: of predicted positives, how many were actually positive – important when false positives are costly.
- **Recall**: of actual positives, how many were correctly identified – critical in medical diagnosis, where missing a malignant case (false negative) is far more dangerous than a false alarm.
- **F1-Score**: harmonic mean of precision and recall, useful when both false positives and false negatives carry meaningful cost.
- **ROC-AUC**: measures the model's ability to discriminate between classes across all classification thresholds, independent of a single cutoff.

For this medical diagnosis context, **recall** is the most important metric: a false negative (missing a malignant tumor) has far greater consequences than a false positive (flagging a benign tumor for further review).

4 Results

Model	Accuracy	F1 Score
Logistic Regression	0.98	0.99
Logistic Regression (Balanced)	0.96	0.96
Decision Tree	0.91	0.93

Table 1: Model comparison on the test set.

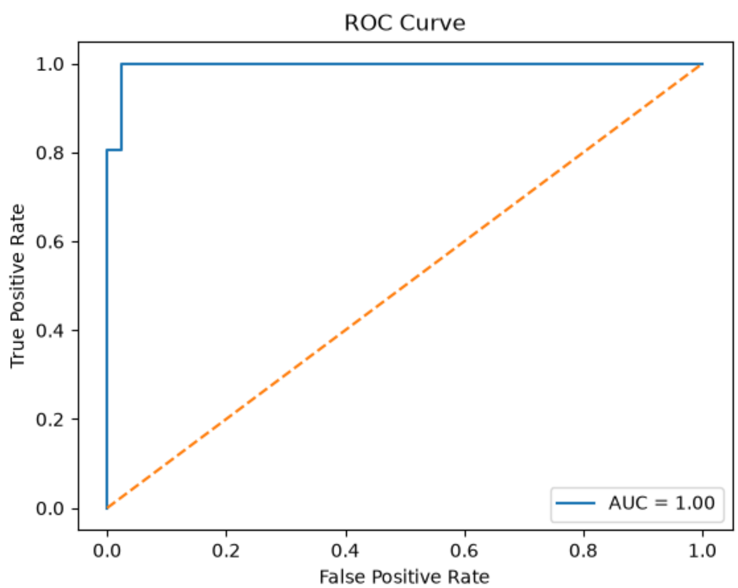

Figure 1: ROC curve for baseline Logistic Regression. AUC = 1.00.

5 Confusion Matrix Analysis

	Predicted Negative	Predicted Positive
Actual Negative	41	1
Actual Positive	1	71

Table 2: Confusion matrix for baseline Logistic Regression.

1 false negative represents a malignant case missed by the model – the most costly error type in this context.

6 Imbalance Handling

Applying `class_weight="balanced"` re-weights the loss function to penalize misclassification of the minority class more heavily. This DID NOT IMPROVE recall on the minority (malignant) class compared to the baseline – recall for the malignant class stayed at 0.98 in both models – while precision for that class dropped from 0.98 to 0.91 and overall F1 fell from 0.98 to 0.94. In this case, the baseline was already well-balanced given the dataset's moderate imbalance, so class weighting introduced a precision trade-off without a corresponding recall gain.

7 Model Comparison

Logistic Regression and Decision Tree were compared on stability and interpretability:

- **Logistic Regression**: More stable, coefficients are directly interpretable, less prone to overfitting on this dataset size.
- **Decision Tree**: Can capture non-linear decision boundaries but is more prone to overfitting without depth constraints, and less interpretable at scale.

8 Final Model Decision

Based on the results, the **baseline Logistic Regression** model is selected as the final model. It achieves the highest accuracy (0.98), F1-score (0.99), and AUC (1.00) among all three models, with only a single false negative and single false positive on the test set. Class weighting did not improve minority-class recall and instead reduced precision, while the Decision Tree underperformed on both metrics. This makes baseline Logistic Regression the most suitable choice for this classification task, offering strong performance without the added complexity or trade-offs introduced by the other approaches.